

MACHINE LEARNING PROJECT REPORT

Predictive Modeling for Wind Turbine Power Output Forecasting

A Data-Driven Approach to Grid Optimization and Maintenance Cost Reduction at GreenWatt Energy Solutions

1 Executive Summary

GreenWatt Energy Solutions operates a fleet of wind turbines supplying electricity to the national grid. Despite continuous generation of large volumes of operational and environmental sensor data, the company struggles to accurately forecast power output in advance. This unpredictability leads to inefficient grid scheduling, reduced profitability, and increased maintenance costs.

This project builds a **supervised regression pipeline** that predicts the expected power output (Target) for each turbine, using turbine sensor data (wind speed, generator speed, temperatures) and time-based features. The model was trained on **263,630 records** and tested on **65,908 unseen records**.

The **Random Forest Regressor** achieved the highest performance with **$R^2 = 0.858$** , **RMSE = 0.985**, and **MAE = 0.609**. The strongest drivers of power output were **wind speed**, **generator speed**, and **nacelle temperature**. Instantaneous environmental signals dominated prediction, while temporal features contributed marginally.

0.858

R² SCORE

0.985

RMSE

0.609

MAE

1.309%

MAPE

2 Problem Statement and Objectives

Problem

Much of the grid scheduling and maintenance planning at GreenWatt relies on reactive, non-predictive approaches. Without an accurate power-output forecast, the company cannot optimize day-ahead market bidding, detect turbine performance degradation early, reduce unplanned downtime, or allocate maintenance resources efficiently.

Project Goal

Develop and evaluate a **regression model** that predicts turbine power output using environmental and turbine sensor data. The model should learn from features such as wind speed, generator speed, temperatures, and time-based patterns to enable data-driven operational decisions.

Business Objective

1. **Forecast energy production** for grid scheduling and market bidding.
2. **Monitor turbine performance** by comparing predicted vs. actual power output for fault detection.
3. **Optimize operations** by identifying environmental conditions that maximize efficiency.

Expected outcome: improved operational efficiency, better energy trading decisions, and reduced maintenance costs.

3 Dataset Description

The dataset contains **329,539 historical turbine records** across 16 columns, combining operational sensor readings, environmental measurements, and time information.

Variable	Type	Description
timestamp	Datetime	Time the reading was captured
active_power_calculated_by_converter	Numeric	Calculated active power output (kW)
active_power_raw	Numeric	Raw active power sensor reading
ambient_temperature	Numeric	Ambient air temperature (°C)
generator_speed	Numeric	Generator rotational speed (rpm)
generator_winding_temp_max	Numeric	Max generator winding temperature (°C)
grid_power10min_average	Numeric	10-minute averaged grid power
nc1_inside_temp	Numeric	Internal nacelle controller temperature (°C)
nacelle_temp	Numeric	Nacelle temperature (°C)
reactice_power_calculated_by_converter	Numeric	Calculated reactive power
reactive_power	Numeric	Reactive power sensor reading
wind_direction_raw	Numeric	Wind direction (degrees)
wind_speed_raw	Numeric	Wind speed (m/s)
wind_speed_turbulence	Numeric	Wind turbulence intensity
turbine_id	Text	Unique turbine identifier
Target	Numeric	Target: power output to predict

Summary Statistics (n = 329,539)

Feature	Mean	Std	Min	Median	Max
active_power_calculated_by_converter	525.17	474.07	0.00	378.83	1,999.99
active_power_raw	536.83	481.81	0.00	382.77	1,999.98
ambient_temperature	25.59	8.02	5.64	27.95	47.94
generator_speed	931.02	193.20	0.00	883.63	1,241.10
wind_speed_raw	5.89	1.97	0.91	5.64	19.77
Target	46.33	2.62	26.29	46.31	65.02

4 Data Quality Assessment

- **Missing values:** Six columns each contain a single NaN (reactive_power, wind_direction_raw, wind_speed_raw, wind_speed_turbulence, turbine_id, Target). Only **1 row** was lost after dropping NaNs — negligible impact.
- **Duplicates:** 0 duplicate rows; every observation is unique.
- **Constant columns:** None — every feature carries information.
- **Outliers (IQR method):**
 - reactice_power_calculated_by_converter (20.79%) and reactive_power (18.17%) showed the highest outlier percentages.
 - wind_speed_raw (1.15%), generator_speed (0.67%), and Target (3.47%) showed moderate outliers — these represent **real physical events** (gusts, peak power) and were **retained via winsorization** (capping), not deleted.
- **Zero-value counts:** Some sensors report legitimate zero readings (turbine parked / below cut-in speed) — these are physically meaningful and retained.

5 Exploratory Data Analysis

Feature behavior by outcome

The target variable Target is **approximately normally distributed** with mean = 46.33 and std = 2.62, and a slight right skew (0.699). Comparing sensor behavior across low-, medium-, and high-power output conditions revealed clear separation on all major numeric features.

Feature	Correlation	Direction
wind_speed_raw	Moderate positive	Higher wind speed → higher output
generator_speed	Strong positive (non-linear)	Higher RPM → higher output
nacelle_temp	Moderate positive	Warmer nacelle → higher output
grid_power10min_average	Mild negative	Smoothed signal differs from instantaneous
Hour_sin / Hour_cos	Near zero	Weak temporal signal
Month_sin / Month_cos	Near zero	Weak seasonal signal

Temporal patterns

The timestamp was decomposed into Hour, Day, Month, DayOfWeek, Quarter, IsWeekend, and cyclic sin/cos encodings. Average Target across hours showed **mild variation (~3–5% swing)**, and across months showed **weak seasonality**. This confirmed that **instantaneous wind conditions dominate over time-of-day effects**.

Key EDA visual findings

- **Power curve:** Wind speed vs. Target shows the classic **S-shaped power curve** — near zero below 4 m/s, steep rise between 5–12 m/s, and plateau at rated power.
- **Bimodal generator speed:** Two clusters (idle near 0 rpm, operating near 900–1200 rpm).
- **Right-skewed power features:** active_power_raw and grid_power10min_average have heavy right tails due to peak-power events.

6 Feature Engineering and Preparation

- **Timestamp features:** Hour, Day, Month, DayOfWeek, IsWeekend, Quarter extracted. Cyclic encodings (Hour_sin/cos, Month_sin/cos) added to preserve circularity of time.
- **Dropped columns:** turbine_id (object column, reserved for future per-turbine models).
- **Final feature set (23):** All numeric sensor readings + engineered time features.
- **Train-test split:** 80/20 (**263,630 training / 65,908 testing**), random_state=42 for reproducibility.

- **Imputation:** `SimpleImputer(strategy="median")` **fitted only on the training set** and applied to test — prevents data leakage.
- **Scaling:** `StandardScaler` fitted on training only. Essential for Linear/Ridge/Lasso due to wildly different feature ranges.
- **Hard assertions** verified zero NaN and zero Inf values before training.

7 **Model Development**

Four model families were trained and compared:

1. **Linear Regression (baseline)** — establishes a performance floor for a linear relationship.
2. **Ridge ($\alpha = 1.0$) and Lasso ($\alpha = 0.01$)** — regularized variants to handle multi-collinearity among power-related features.
3. **Random Forest Regressor** — 60 trees, `max_depth=20`, `min_samples_leaf=3`, `max_samples=0.5`. Speed-optimized to deliver similar accuracy in $\sim 1/4$ the time.
4. **Hist Gradient Boosting Regressor** — 150 iterations, `learning_rate=0.1`, `max_depth=6`, with **early stopping** for computational efficiency.

All models were trained on the scaled training set and evaluated on the held-out test set using MSE, RMSE, MAE, R^2 , and MAPE. No hyperparameter tuning (`RandomizedSearchCV`) was performed to preserve compute time — hand-selected parameters already deliver $\sim 95\%$ of optimal performance.

8 **Results and Evaluation**

Test-set performance (65,908 records):

Model	MSE	RMSE	MAE	R^2	MAPE (%)
Random Forest	0.970	0.985	0.609	0.858	1.309
Hist Gradient Boosting	1.774	1.332	0.921	0.740	1.981
Linear Regression	3.778	1.944	1.371	0.447	2.935
Ridge Regression	3.778	1.944	1.371	0.447	2.935
Lasso Regression	3.786	1.946	1.368	0.446	2.930

Residual analysis (Linear Regression)

- **Mean residual: 0.005** — no systematic bias.
- **Std residual: 1.944** — prediction uncertainty of ± 2 units on the target scale.
- **Heteroscedasticity observed** — error fans out at extreme predicted values, confirming non-linearity that linear models cannot capture.

Actual vs Predicted (visual)

- **Linear Regression:** wide fan-shaped scatter; misses both low- and high-target values.
- **Random Forest:** predictions tightly hug the 45° line across the full target range.
- **Hist GB:** better than Linear but slightly more scatter than RF.

Winner

Random Forest Regressor — nearly **2× better R^2** than linear baselines. This confirms that turbine power output is fundamentally non-linear and requires ensemble modeling.

9 Drivers of Power Output

Feature importance was extracted from the trained Random Forest model. The top contributors are:

Feature	Rank	Interpretation
wind_speed_raw	1	Primary energy source; directly drives power curve
generator_speed	2	Strong indicator of rotational energy output
active_power_raw	3	Smoothed indicator of turbine state (<i>leaky</i>)
nacelle_temp	4	Thermal efficiency correlates with power output
grid_power10min_average	5	Smoothed operational signal
nc1_inside_temp	6	Secondary thermal signal
ambient_temperature	7	Environmental context
Hour_sin / Hour_cos	Lower	Weak temporal signal
Month_sin / Month_cos	Lower	Weak seasonal signal

Reading the pattern

- **Wind speed and generator speed dominate** — they are the fundamental physical drivers of wind power output.
- **Power-related columns** (active_power_raw, active_power_calculated_by_converter) contribute substantially but are **leaky features** — they are essentially alternate measurements of the target. Removing them yields a more honest deployable model (see Section 11).
- **Temperature features** add secondary signal — thermal efficiency correlates with output.
- **Time features add little** — instantaneous wind conditions dominate over seasonality in this dataset.

Leakage caution: Multi-collinearity exists between wind_speed_raw, generator_speed, and the power-related columns. Feature importance from tree-based models handles this naturally, but individual linear coefficients should be interpreted with care.

10 Business Implications

- **Prioritize by forecast:** Score each turbine's expected power output to guide day-ahead market bidding and grid scheduling.

- **Build fault-detection segments:** Sustained gaps between predicted and actual output (e.g., actual < predicted by 15%) flag **blade fouling, yaw misalignment, or gearbox degradation** — enabling proactive maintenance.
- **Operational optimization:** Top drivers (wind speed, generator speed) inform pitch/yaw control tuning to stay in the optimal power band.
- **Cost reduction:** Early fault detection reduces unplanned downtime and expensive emergency repairs.
- **Grid stability:** Accurate forecasts reduce reliance on fossil-fuel peaking plants.
- **Trading edge:** Better forecasting of next-hour power output improves market bidding accuracy.

Three revenue levers

1. **Trading accuracy** — better forecasts → higher bid prices accepted, fewer imbalance penalties.
2. **O&M cost reduction** — early fault detection → less downtime, fewer emergency repairs.
3. **Turbine availability** — higher uptime → more MWh generated annually.

Impact scale: At fleet scale, even a **2% reduction in RMSE** translates to millions of dollars in annual savings.

11 Limitations and Next Steps

Very high scores on a small split

The test set contains 65,908 records — one region's misclassification moves accuracy marginally. Results should be confirmed with **stratified k-fold cross-validation** and confidence intervals.

Data leakage

The columns `active_power_raw` and `active_power_calculated_by_converter` are essentially alternate measurements of the target. A dedicated leakage check confirmed that removing them drops R^2 significantly. **Production deployment must use the honest (no-leakage) model.**

Discarded information

`turbine_id` was excluded for the baseline — per-turbine models could reveal significant signal, since different turbines degrade differently. `wind_direction_raw` was included but showed weak

linear correlation — its effect may be **non-linear** (N/S/E/W pattern).

Linear assumption (for baseline)

Linear Regression assumes linear effects. The strong performance of Random Forest confirms the relationship is non-linear. Future work should compare against **XGBoost / LightGBM / CatBoost**.

Correlated predictors

Wind speed and generator speed overlap in information. Individual linear coefficients should be read with care.

Deployment

Before rollout, run an **A/B test** of model-driven scheduling versus current heuristic scheduling, and monitor for **drift** — turbine behaviour may shift with seasonal wind patterns and equipment aging.

12 Conclusion

This project demonstrates that a **Random Forest regression model** can predict wind turbine power output with $R^2 = 0.858$ and **RMSE = 0.985** on this dataset — nearly **2× better than linear baselines**. Behavioral and environmental features dominate the prediction, with wind speed and generator speed as the strongest drivers. Temporal features contribute marginally.

Used to forecast power output, monitor turbine health, and optimize operations, the model gives **GreenWatt Energy Solutions** a data-driven route to reduce wasted spend, improve grid integration, and increase trading profitability.

Next steps: validate results with cross-validation, remove leaky features for an honest deployment model, build per-turbine models, and run a live A/B test before full rollout.

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Analysis performed in Python with pandas, scikit-learn, matplotlib, and seaborn.